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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (CST) (CBCS) Degree Examinations

May- June 2025

22CST202 – FINITE ELEMENT METHOD OF ANALYSIS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

Unit – I		Marks	BT*	CO*	PO*
1. a)	Discuss the advantage and disadvantage of finite element over classical methods.	10	L*2	1	2
b)	Distinguish between the Finite element method and finite difference method.	10	L2	1	2
2. a)	Explain the principle and steps involved in the Rayleigh-Ritz method.	10	L2	1	2
b)	Demonstrate the various elemental shape used in finite element analysis with neat sketches and suitable examples for applicability.	10	L2	1	2
Unit – II					
3. a)	State and explain the convergence requirement of polynomial function.	10	L2	2	2
b)	Illustrate the geometric isotropy/geometric invariance. Why should polynomial function satisfy these requirements?	10	L2	2	2
4. a)	Determine the shape function for two noded beam element using generalized coordinate approach.	10	L3	2	2
b)	Determine the shape function for constant strain triangle(CST) element using polynomial function.	10	L3	2	2
Unit – III					
5. a)	Explain Iso parametric, sub parametric and super parametric elements.	10	L3	3	2
b)	Determine the local coordinated of point 'P' for isoparametric quadrilateral element having a Cartesian coordinate (7,4) as shown in the Fig.Q5(b)				

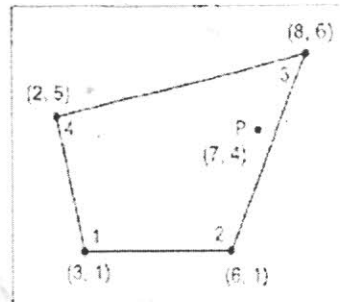


Fig.Q5(b)

10 L3 3 2

6. a) Develop the stiffness matrix for CST element as shown in the Fig.Q6.

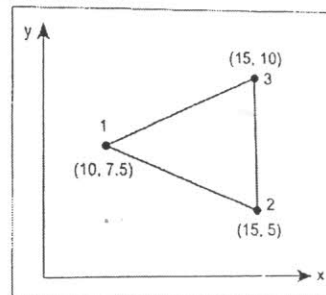


Fig.Q6.

20 L3 3 2

P.T.O

Unit – IV

7. a) Explain the potential energy in elastic bodies. Derive the expression for potential energy in spring.
 b) Assemble equation of equilibrium in matrix form for the spring system as shown in Fig.Q7(b) by potential energy principle.

10 L2 4 2

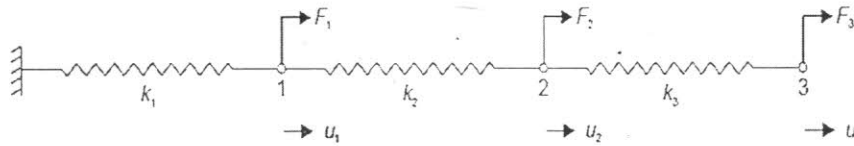
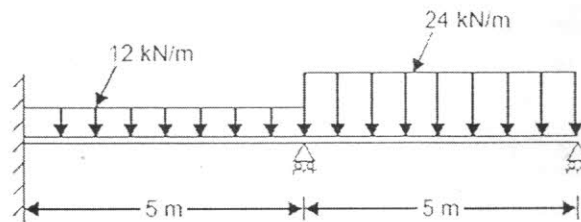


Fig.Q7(b)

10 L3 4 2

8. a) Analyze the beam shown in Fig. Q8 by finite element method and determine the end reactions. Also determine the deflections at mid span. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 5 \times 10^6 \text{ mm}^4$.



20 L3 4 2

Unit – V

9. a) Explain the assumptions and derive the basic relations in bending of thin plate.
 b) Discuss the confirming and non-confirming rectangular plate analysis.
10. a) Explain forces and various factors to be considered in the development of curved shell element.
 b) Write a short note on I) Flat element, II) Curved Element.

10 L2 5 2

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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
